Deggendorf Institute of Technology

Faculty of Applied Natural Sciences and Industrial Engineering

Applied AI for Digital Production Management

Master's Thesis

Entwicklung und Evaluation eines NLP-basierten Konzepts zur strukturierten Extraktion von Informationen und zur Unterstützung der Planung in Instandhaltungsaufträgen

Development and Evaluation of an NLP-Based Concept for Structured Information Extraction and Planning Support in Maintenance Work Orders

Master thesis to obtain an academic degree:

Master of Engineering (M.Eng.)

submitted by

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Submission date: 15.09.2026

Statement of Authorship

I hereby declare that I have written the present master's thesis independently and without the use of any sources or aids other than those indicated. All passages taken verbatim or in substance from published or unpublished works of others have been identified as such. This thesis has not been submitted, either in the same or a similar form, to any other examination authority, and has not yet been published.

Deggendorf, 15.09.2026

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Abstract

Aviation maintenance work orders record valuable hands-on experience, yet abbreviations, inconsistent spelling, and short free-text fields make that knowledge hard to search and reuse. This thesis develops and evaluates AviMaint-DSS, an evidence-grounded natural language processing pipeline that converts noisy problem–action records into structured, searchable historical cases. The approach combines controlled text normalization, a compact nine-entity, eleven-relation extraction schema, a comparison of classical, recurrent, and transformer-based extractors, an external benchmark for portability, and a leakage-safe retrieval interface. Byte-level neural normalization reproduced the expert-cleaned reference most faithfully (94% exact match), and joint span-based extraction achieved the strongest relations (strict micro-F1 0.85), while classical models stayed competitive for entities. Under leakage-controlled evaluation, structured retrieval recovered the recorded action family for roughly 80% of queries at rank one and 95% within the top three, with calibrated confidence. The central contribution is an auditable planning-support concept that preserves provenance, exposes alternatives and uncertainty, and leaves maintenance authority with qualified personnel.

Keywords: aviation maintenance; natural language processing; text normalization; information extraction; relation extraction; case retrieval; decision support

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Index of Abbreviations

▸ *Alphabetical. Define each abbreviation in full at its first use in the text as well. Do not list everyday abbreviations (e.g., etc.).*

Abbreviation	Meaning
AI	Artificial Intelligence
CER	Character Error Rate
CRF	Conditional Random Field
DSS	Decision-Support System
ECE	Expected Calibration Error
IE	Information Extraction
KG	Knowledge Graph
MRR	Mean Reciprocal Rank
NER	Named-Entity Recognition
RE	Relation Extraction
RQ	Research Question
SpERT	Span-based Entity and Relation Transformer
WER	Word Error Rate

List of Symbols

▸ *Optional but recommended if you use equations. Define each symbol at first use in the text too.*

Symbol	Meaning
P	Precision (strict, micro)
R	Recall (strict, micro)

Symbol	Meaning
F1	Harmonic mean of precision and recall
n	Number of evaluable queries / records

1 Introduction

1.1 Context and Motivation

Maintenance work orders are more than administrative records. Each entry documents a problem that was reported and an action that was taken, and across thousands of entries these short notes accumulate into an organisational memory of how equipment fails and how those failures were handled. For a planner facing a new fault, that memory is the most directly relevant evidence available: it records what colleagues did, on the same equipment, under comparable circumstances. Making it searchable would shorten the time spent rediscovering knowledge the organisation already possesses.

The obstacle is the form in which knowledge is stored. Work orders are written quickly, for a local reader, and with the immediate goal of closing a task rather than informing a future search. They are therefore concise, context-dependent, and rarely composed as complete sentences. Brundage et al. [1] argue that general-purpose natural language processing pipelines are not sufficient in this setting and propose technical language processing, a domain-driven approach in which engineering use cases, dictionaries, annotated data, and domain experts are treated as parts of one system rather than as afterthoughts.

The specific corpus studied here illustrates the difficulty precisely. The aviation portion of the MaintNet collection [2, 3] contains 6,169 paired problem and action records. The problem field is limited to sixty characters and the action field to fifty-five, and many records run to those limits, so information is sometimes physically cut off in the source. Within those short fields the text contains local abbreviations, inconsistent spelling, irregular punctuation, and telegraphic grammar. Two records describing the same technical situation may therefore share almost no identical words, which prevents them from being matched by any method that relies on surface form.

Three research resources make the problem tractable. MaintNorm [4] established a corpus and benchmark for cleaning industrial maintenance short text. MaintIE [5] demonstrated that maintenance text can be represented as typed entities and directed relations, and supplied an expert-annotated benchmark. Byte-level language models such as ByT5 [6] process text as raw bytes rather than fixed sub-word units, which makes them robust to exactly the spelling-level noise that dominates this corpus. Each is a valuable component, but none of them delivers a

connected process from raw text through structure to retrievable, traceable evidence that a planner can actually use.

1.2 Problem statement

The current situation presents four connected problems. First, surface noise prevents records describing the same technical concept from matching one another. Second, the information that matters is embedded in unstructured text, so a system cannot distinguish an item from a fault, an action, a location, an observation, an outcome, or a reference. Third, any structure predicted automatically across a full corpus is model output rather than verified truth and treating it as truth would misrepresent its reliability. Fourth, a superficially similar historical problem may record an action that was unsuccessful, incomplete, deferred, or inapplicable to the new situation, so similarity alone is a poor guide.

The target state is an auditable pipeline that preserves the raw text, extracts typed structure, builds traceable cases, and retrieves evidence for human review while communicating how far that evidence should be trusted. A central question is whether the most complex available model is required. This thesis therefore compares three families of extraction methods under identical conditions: a classical feature-based approach, a recurrent neural approach, and Transformer-based approaches in both pipeline and joint form. That controlled comparison, rather than the adoption of any single model, is the core of the work.

1.3 Aim, objectives, and research questions

The aim of this thesis is to develop and evaluate an auditable, NLP-based concept that converts noisy aviation maintenance work orders into structured historical evidence and uses that evidence to support, but not replace, human maintenance planning.

The objectives are to compare controlled text-normalization strategies against an expert-cleaned reference; to define a compact entity and relation schema and construct a manually corrected gold corpus; to compare four extraction models under one frozen evaluation; to test the pipeline on an independent maintenance benchmark; to convert the full corpus into a traceable structured layer; and to implement and evaluate a planning interface that presents evidence, alternatives, provenance, and uncertainty. These objectives are addressed through five research questions.

- RQ1 - Normalization. To what extent can controlled text normalization reduce noise in aviation maintenance work orders while preserving technical meaning, and does the cleaner representation improve downstream information extraction and case retrieval?
- RQ2 - Structured representation. Which entity and relation schema represents these work orders well enough for planning support while remaining learnable from the available annotations?
- RQ3 - Information extraction. Which modelling approach extracts maintenance entities and their relations most accurately and robustly from short, noisy work orders?
- RQ4 - Leakage-safe retrieval. Which representation retrieves the most useful historical cases when only the problem description is used as the query and the recorded action remains hidden until after retrieval?
- RQ5 - Planning support. How can retrieved evidence, provenance, alternatives, negative outcomes, and uncertainty be presented as useful planning support without appearing to approve a maintenance procedure?

Table 1.1 states each research question together with the population it is evaluated on and the strength of evidence that population can carry. The distinction is maintained throughout the thesis: a figure obtained on development data is never presented as a performance estimate, and structure predicted over the full corpus is never presented as verified truth. Figure 1.1 places the same five questions on the processing chain they belong to.

RQ	What it asks	Evaluation population	Evidence level
RQ1	Whether controlled cleaning reduces noise without changing meaning, and whether it helps downstream	741 held-out normalization records; 225 extraction test records	Reference
RQ2	Whether a compact typed schema is both expressive enough and learnable	225 extraction test records; 6,169 full-corpus records	Reference and silver
RQ3	Which modelling approach extracts entities and relations most accurately	225 frozen test records; 108 MaintIE test records	Locked test
RQ4	Which representation retrieves the most useful historical cases without leakage	724 evaluable locked-test queries	Locked test

RQ5	How evidence and uncertainty can be presented as planning support	Development calibration and locked test	Locked test

Table 1.1. The five research questions, the population each is answered on, and the strength of evidence that population supports.

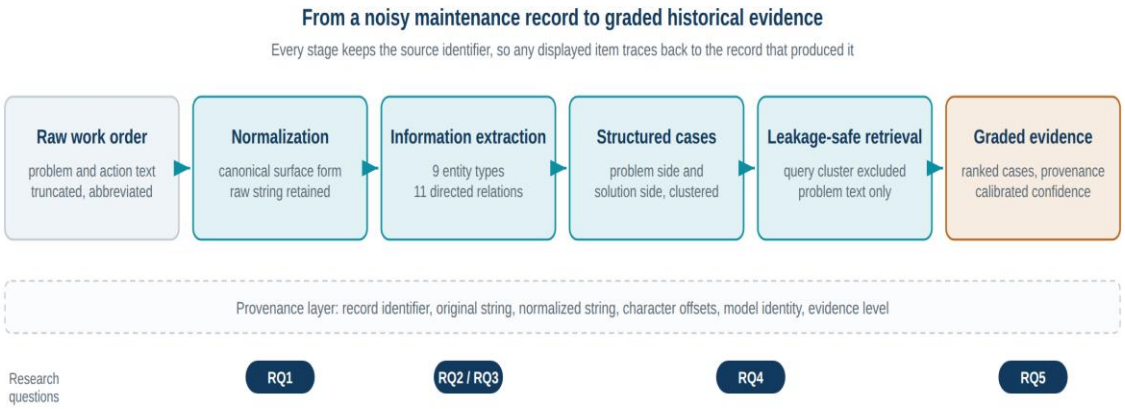


Figure 1.1. The processing chain examined in this thesis, from a raw work order to graded historical evidence, with the research questions positioned at the stages they address.

1.4 Scope and safety boundary

The scope is English aviation problem and action text from MaintNet, covering normalization, annotation, entity and relation extraction, structured case construction, retrieval, ranking, evidence presentation, and a local planning dashboard. MaintIE [5] is used solely as an external benchmark of portability. Generative large language models are deliberately excluded from the decision path, because a model that composes fluent new text cannot guarantee that every displayed statement is grounded in a real historical record.

A safety boundary shapes the entire design. Historical evidence and maintenance authority are distinct. A previous action shows what was once done; it does not establish that the same action is correct for another aircraft, configuration, or operating context. Aviation regulation requires current, applicable maintenance data: in the United States, 14 CFR 43.13 requires the use of methods, techniques, and practices prescribed in current manufacturer instructions or other methods acceptable to the Administrator [7], while in Europe Commission Regulation (EU) No 1321/2014 requires applicable current maintenance data to be held and used [8]. The system studied here therefore organizes and displays past records; it does not determine airworthiness, approve work, select a regulatory basis, confirm applicability, calculate failure rates, or release

an aircraft to service. This position is consistent with the emphasis on explainability, learning assurance, and human oversight in the European Union Aviation Safety Agency's guidance for machine-learning applications [9], and with the risk-management principles of validity, transparency, accountability, and human oversight set out in the NIST AI Risk Management Framework [10].

1.5 Contributions

The thesis makes five contributions. It compares four normalization conditions and shows that intrinsic agreement with a cleaning reference and downstream extraction quality do not favour the same method. It defines a nine-entity, eleven-relation schema and constructs a 1,600-record manually corrected gold corpus with frozen evaluation partitions. It provides a controlled comparison of classical, recurrent, and Transformer extraction models on that frozen data, together with an independent test on MaintIE. It converts all 6,169 records into a traceable structured layer that is explicitly labelled as model output rather than verified truth. Finally, it implements and evaluates a planning interface that presents complete historical cases, alternatives, negative evidence, provenance, calibrated uncertainty, and graded confidence, and that abstains when evidence is absent. The overall achievement is a reproducible and auditable path from noisy records to searchable historical evidence for human planning review.

1.6 Structure of the thesis

Chapter 2 reviews the scientific background and related work on maintenance language, lexical normalization, information extraction, case retrieval, uncertainty, and auditable decision support, and identifies the research gap. Chapter 3 documents the data sources, the evidence hierarchy, the extraction schema, the construction of the gold corpus, and the experimental design. Chapter 4 explains the methodology and the theory behind each method, including the evaluation metrics. Chapter 5 reports the measured results. Chapter 6 discusses those results against the research questions, compares them with prior work, and examines threats to validity. Chapter 7 concludes, states the contributions against the objectives, and defines the future evaluation path.

2 Fundamentals

▸ 8–16 pages. Explain all background needed to follow the methodology: maintenance work-order data, text normalization, information extraction (NER/RE, SpERT), case-based retrieval, calibration. Define key terms and delineate related work.

2.1 Maintenance Work Orders and the MaintNet Corpus

[…]

2.2 Text Normalization

[…]

2.3 Information Extraction: Entities and Relations

[…]

2.4 Case Retrieval and Uncertainty Calibration

[…]

2.5 Related Work

[…]

3 Methodology

▸ 2–4 chapters, ~10–20 pages each. Describe the complete solution and justify every choice (why method A over B). Clearly separate your own work from prior work. Each subsection below corresponds to one research question.

3.1 RQ1 — Text Normalization

▸ Dataset & cluster-safe split (5,124 primary → 3,457 / 926 / 741; 1,045 sensitivity; seed 42), four conditions (raw / rules / ByT5 / hybrid), evaluation design.

[…]

3.2 RQ2 — Structured Representation

[…]

3.3 RQ3 — Information-Extraction Models

[…]

3.4 RQ4 — Leakage-Safe Case Retrieval

[…]

3.5 RQ5 — Calibrated Planning Support

[…]

4 Results

‣ 4–12 pages. Present and explain results only — do NOT interpret or discuss here. Give individual results, then summarise the most important ones in a clearly structured table. Every figure/table must be referred to in the running text. Report deviations/inaccuracies.

[…]

5 Discussion

‣ Interpret the results against each research question, with the claim boundaries (historical-agreement ≠ technical correctness). Compare against the state of the art from the Fundamentals chapter.

[…]

6 Summary and Prospects

‣ 1–3 pages. Restate the research question, approach, key results and their evaluation; answer the questions from the Introduction; reflect critically (what worked, what could be better); give an outlook. Introduce no new ideas here.

[…]

7 Bibliography

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[6] Xue, L.; Barua, A.; Constant, N.; Al-Rfou, R.; Narang, S.; Kale, M.; Roberts, A.; Raffel, C.: ByT5: Towards a Token-Free Future with Pre-trained Byte-to-Byte Models. In: Transactions of the ACL, Vol. 10, 2022, pp. 291-306.

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[8] European Commission: Commission Regulation (EU) No 1321/2014 on the continuing airworthiness of aircraft and aeronautical products, parts and appliances. Official Journal of the European Union, 2014.

[9] European Union Aviation Safety Agency (EASA): EASA Concept Paper - Guidance for Level 1 and 2 Machine Learning Applications. Cologne, 2024.

[10] National Institute of Standards and Technology: Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST, Gaithersburg, 2023.

8 Appendix

▸ *Own title page in the final version. Number items A1, A1.1, A2 … and list them in the Table of Contents. Programming code goes on the electronic data carrier — it is NOT printed.*

8.1 A1 Additional Result Tables

[…]

8.2 A2 Full Schema Definition

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